

Chenrui Zhou

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Education

Johns Hopkins University, Whiting School of Engineering — MSE in Computer Science (Jan 2024 – May 2025)

Nanjing University, Kuang Yaming Honors School — B.S. in Information and Computational Science (Mathematics) (Sep 2019 – Jun 2023)

Technical Skills

Programming: Python, C/C++, SQL, MATLAB

Frameworks: PyTorch, TensorFlow, OpenCV

Areas: Computer Vision, Diffusion Models, 3D Vision, Reinforcement Learning

Tools: Linux, Git, Data Processing, Model Training & Evaluation

Project Experience

Scaling-up NeMo 3D Pose Estimation Model — Johns Hopkins University (CCVL Lab) (May 2024 – Mar 2025)

- Expanded a 3D pose estimation model to large-scale multi-category datasets.
- Modified training pipelines to support joint training across heterogeneous object categories.
- Implemented and evaluated the complete training and testing pipeline in PyTorch.
- Verified cross-category generalization ability for pose estimation tasks.

P2P-3DEdit: Prompt-to-Prompt 3D Editing via NeRF — Nanjing University (Jun 2023 – Jan 2024)

- Developed a pipeline for controllable 3D editing using diffusion-based prompt manipulation.
- Integrated Prompt-to-Prompt attention control into NeRF-based 3D generation frameworks.
- Built interfaces between NeRF models and diffusion backbones including Stable Diffusion and SDXL.
- Evaluated generation quality and editing controllability through large-scale experiments.

Robustness of Prompt-Reward Entanglement in Reinforcement Learning — National University of Singapore (Jul 2025 – Present)

- Investigating robustness issues caused by prompt-reward coupling in RL training.
- Conducting theoretical analysis of model generalization under reward manipulation.
- Developing mathematical explanations for structural generalization degradation.

Quantitative Multi-Factor Stock Strategy Backtest (Independent Project)

- Implemented a multi-factor equity trading strategy using valuation and momentum signals.
- Factors include PE ratio, PB ratio, and price momentum.
- Built a Python backtesting framework to evaluate Sharpe ratio, drawdown, and IC metrics.

Awards

Meritorious Winner — Mathematical Contest in Modeling (MCM/ICM), 2021

Outstanding Graduate — Kuang Yaming Honors School, Nanjing University, 2023